LAB 1 – INTELLIGENT DECISION-MAKING IN ROBOTICS

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In this practice I didn't need to use any AI tool, as the codes and slides provided by the professor were enough to understand all the concepts of the Behavior Trees.

Exercise 2 and 3:

This tree consists of a selector that leads to two sequences, one of them is for the situation where the robot is already in the location and the other one if the robot needs to be moved to the location.

Exercise 4:

The problem with the code was the fact that the parallel node was the root. To solve this problem, I changed the root to a sequence node whose children are a parallel node with the battery and obstacles check and then the move forward operation. The tree can be seen in the "Trees" pdf.

Exercise 5:

If the failure node returns RUNNING the inverter cannot do his job because the state RUNNING does not have an invert state.

```
(venv) ignagonmell@DESKTOP-SVHOVNC:~$ python EXERCISE5.py
Returning FAILURE, but Inverter will flip it
-^- Succeed after fail [*]
    --> Always Fail [*]
```

Exercise 6:

In this exercise the fourth interaction acts as a first one.

```
(venv) ignagonmell@DESKTOP-SVHOVNC:~$ python EXERCISE6.py
Tick 1
Always failing
-^- three times retry [*] -- attempt failed [status: 1 failure from 3]
    --> Always Fail [✕]

Tick 2
Always failing
-^- three times retry [*] -- attempt failed [status: 2 failure from 3]
    --> Always Fail [✕]

Tick 3
Always failing
-^- three times retry [✕] -- final failure [status: 3 failure from 3]
    --> Always Fail [✕]

Tick 4
Always failing
-^- three times retry [*] -- attempt failed [status: 1 failure from 3]
    --> Always Fail [✕]
```

Exercise 7:

I changed the “ON_SUCCESFUL_COMPLETION” policy to “ON_COMPLETION”. This means that the message will be printed even if the node returns FAILURE.

```
(venv) ignagonmell@DESKTOP-SVHOVNC:~$ python EXERCISE7.py
Message printed only once: SUCCESS
^C(venv) ignagonmell@DESKTOP-SVHOVNC:~$ python EXERCISE7.py
Message printed only once: SUCCESS
^C(venv) ignagonmell@DESKTOP-SVHOVNC:~$ python EXERCISE7.py
Message printed only once: FAILURE
```

Final Exercise:

The structure of the exercise consists of a sequence root whose first child is a decorator node to retry the human detector function three times. Then the code executes entirely except if the human doesn’t need help, in that case the robot restarts his execution.